Symmetry-Compatible Hypothesis Volume Predicts Out-of-Distribution Generalization

Subtitle. Weakness, not generic compression alone, in shortcut-compatible learning problems.

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Abstract

We study model selection in finite-domain settings where the training data admits both a local shortcut and a globally symmetry-compatible rule. In these settings, training loss and in-distribution validation cannot distinguish the two completions, and generic simplicity or flatness proxies can prefer the shortcut. We define a finite operationalization of Bennett-style **weakness** as the number of transformations under which a candidate function remains equivariant up to an output action. Across cyclic and dihedral symbolic task families (500 trials each), weakness selects the invariant rule in 100% of trials (Wilson 95% lower bound 0.992), while the tested train-loss, validation, description-length, MDL-style, compression, and flatness proxies select the shortcut in 100% of trials (Wilson 95% upper bound 0.008). A data-inferred selector, built by enumerating circular-domain transformations and retaining those consistent with training pairs, matches the oracle selector on cyclic and dihedral families without using the hidden offset or reflection parameter. On trained neural models, learned-function weakness under the true group is the strongest correlate of OOD accuracy across 256 local MLPs (Pearson r = +0.817, Fisher 95% CI +0.772 to +0.854), a 1024-model Modal resample (r = +0.813), and a 4096-model Modal rescale (r = +0.8085, CI +0.798 to +0.819). A 4096-model augmentation-fixed-effect check leaves a positive residual correlation (r = +0.488, CI +0.465 to +0.511), so the effect is not only an augmentation-condition label. A rotated-stroke vision benchmark gives the same ordering (rotation weakness r = +0.672; wrong-group control r = -0.341). Parity is a clean negative case, and S_n is a large-group boundary case: oracle weakness helps, but data-inferred group discovery degrades. An appendix reports a Pythia-70M paraphrase latent-geometry check, not behavioral OOD evidence. The claim is not that compression is useless in general; it is that, in shortcut-compatible symmetry tasks, the relevant inductive bias is symmetry-compatible hypothesis volume.

1. Introduction

Modern learning pipelines are often underspecified: many predictors achieve indistinguishable training or in-distribution validation performance, yet behave differently under deployment shifts. Shortcut learning is one familiar manifestation of this problem: a local rule solves the observed benchmark but fails when the deployment distribution changes. In the experiments below, we make this ambiguity explicit. The training data is compatible with both a local shortcut and a globally transportable rule. The question is which train-perfect completion a selector should prefer.

The usual model-selection vocabulary gives several candidates: minimize training loss, prefer shorter descriptions, approximate Solomonoff or MDL priors, prefer flat minima, minimize parameter norm, or trust held-out validation. These are valuable heuristics in many settings, but they are not targeted at the particular ambiguity created by partially observed symmetries. If the deployment distribution is generated by a transformation structure, the relevant question is not only "which hypothesis is short?" but "which hypothesis remains valid under the transformations that generate the missing cases?"

This paper tests that question through **weakness**: the size of the transformation set under which a candidate function remains compatible with itself. The term comes from Bennett's compatible-world-volume framing, but the operational measure here is deliberately finite and checkable: count the group elements for which the learned function is equivariant up to an output-side action.

Contributions:

1. We define a finite-domain weakness score for candidate functions and learned neural function tables.
2. We construct shortcut-compatible symbolic tasks where the local shortcut and the invariant rule are both train-perfect, but only the invariant generalizes to OOD inputs.
3. We show that weakness selects the OOD-correct completion on cyclic and dihedral families where the candidate group has the right granularity, while the tested loss, validation, description-length, compression, MDL-style, and flatness proxies do not.
4. We verify the neural version across 256 local MLPs, a 1024-model Modal replication, and a 4096-model Modal rescale.
5. We add a methods appendix with exact task definitions, selector definitions, data-inferred group pseudocode, neural training distributions, OOD splits, correlation intervals, and an augmentation-fixed-effect check.
6. We report honest boundaries: parity is too small a group to disambiguate, the full symmetric group is too uninformative for the current data-inferred heuristic, and the language result currently supports only a latent-geometry claim.

Claim level. This is evidence for a symmetry-volume model-selection principle in finite shortcut-compatible regimes. It is not a claim that generic compression, PAC-Bayes, validation, or flatness never matter. A safer summary is: when train-perfect shortcuts and invariant rules coexist, generic simplicity proxies can be underdetermined or misaligned, while symmetry-compatible volume directly measures the completion that will transport across the group-generated OOD split.

2. Definitions and Selection Principle

Let $f: X \to X$ be a candidate function on a finite domain X of size n. Let G be a finite group or finite transformation set acting on X.

With-action equivariance. We say f is compatible with $g \in G$ if there exists an output-side action $h \in G$ such that

$$f(g \cdot x) = h \cdot f(x) \ \forall x \in X.$$

This generalizes strict equivariance ($h = g$) and is useful when input and output actions are related but not necessarily identical.

Weakness.

$$W\_G(f) = \left| g \in G : \exists h \in G, \forall x, f(g \cdot x) = h \cdot f(x) \right|.$$

A function with $W\_G(f) = |G|$ is fully compatible with the candidate transformation set. A function with $W\_G(f) = 1$ is compatible only with the identity.

Weakness selector. Given a finite candidate pool f_1, ..., f_K and a candidate transformation set $\hat{G}$, the weakness selector returns

$$f\_i^{\wedge}*, i^{\wedge}* \in \mathrm{argmax}\_i \ W\_\hat{G}(f\_i),$$

with ties broken by the same compression proxy used by the baseline selectors.

Operational selection principle. If the training data constrains only part of a transformation orbit, then a train-consistent hypothesis that remains compatible with more elements of the deployment-generating transformation set should cover more of the missing orbit. The experiments below test this principle in regimes where the shortcut and invariant are both train-perfect, so the comparison is not confounded by ordinary training error.

The principle has a clear failure mode. If Ĝ is too small, many wrong local completions can be equally compatible. If Ĝ is too large or semantically uninformative, wrong candidates may have comparable compatibility counts. Those are not footnotes; they are part of the proposed operating regime.

Proposition 1 (group-completed coverage). Suppose deployment inputs are generated by applying a subset S subseteq G of transformations to observed training inputs, and suppose labels are generated by an output action in the same transformation family. Among train-consistent hypotheses with zero compatibility error on observed orbits, a candidate compatible with more elements of S correctly transports more observed labels to the group-completed deployment set. Thus, when S is unknown but contained in the candidate family, W_G(f) is an upper-level proxy for missing-orbit coverage. The proposition is modest: it does not say weakness helps when G is misaligned, too small, or too large to distinguish the competing train-consistent functions.

3. Symbolic Benchmark

3.1 Task Families

We use four families. Each trial contains a finite domain, a ground-truth function, a biased training subset, an OOD subset, and a candidate pool containing at least a local shortcut and the true invariant.

- **`cyclic_prefix_shift`**: domain Z_n, n ∈ 7, 11, 13. Truth f(x) = (x + b) mod n. Training observes a prefix window w ∈ 3,4,5; OOD inputs are the suffix.
- **`dihedral_reflection`**: domain Z_n, n ∈ 7, 9, 11, 13. Truth f(x) = (b - x) mod n. Training observes a prefix window; OOD inputs are the remaining indices. The candidate transformation family is D_n.
- **`parity_coset`**: even domain size n ∈ 6,8,10. Truth swaps each parity partner x ↦ x ⊕ 1. Training observes one parity coset; the other coset is OOD.
- **`color_permutation`**: domain size n ∈ 4,5,6. Truth is a sampled non-identity permutation. Training observes a sparse subset. The candidate transformation family is S_n.

The candidate pool always includes a local patch that copies the observed training outputs and defaults to identity elsewhere, a memorizer, the true invariant, and available wrong invariants that also fit the training pairs. The local patch is intentionally short; this makes the compression comparison adversarial rather than decorative.

3.2 Selectors and Baselines

We compare eleven selectors. In the discussion below, "classical baselines" means these implemented proxies, not every possible MDL, PAC-Bayes, flatness, or validation method. Appendix A.2 gives the full score/tie-break/access table.

Selector	Definition
`train_loss`	best training accuracy, ties broken by simplicity
`validation`	leave-one-out accuracy on observed training pairs
`simplicity`	shortest hand-coded hypothesis form length

`compression`	`form_length + 20 * train_errors`
`mdl_program`	2^-form_length, a Solomonoff-style proxy
`flatness_proxy`	count of domain positions unconstrained by training
`weakness_oracle`	maximum weakness under the true candidate group
`weakness_wrong_group`	weakness under random non-cyclic permutations
`weakness_noisy_group`	weakness under a 50%-corrupted subset of the true group
`weakness_data_inferred`	weakness under transformations retained by training-pair consistency
`random`	uniform random train-consistent candidate

The validation baseline is deliberately in-distribution: it holds out one observed training pair and scores the fixed candidate on that pair. Validation examples are drawn from the same shortcut-compatible partial-orbit distribution as training; no held-out OOD transformations are included. Wilson intervals use the standard score interval with z = 1.959963984540054.

The `weakness_data_inferred` selector does not receive the hidden offset, reflection parameter, or ground-truth function. For cyclic tasks it enumerates circular shifts and keeps shifts that do not contradict observed pairs under some output shift. For dihedral tasks it enumerates the signed circular prior (rotations plus reflections) and keeps transformations consistent with the observed pairs under some output-side signed action. For color permutation, the current heuristic falls back to a small cyclic prior and is intentionally weak.

3.3 Results

Each family uses 500 independent trials. Wilson 95% intervals are shown for invariant-recovery rates.

`cyclic_prefix_shift`

Selector	Invariant rate	Wilson 95% CI	Mean OOD
`weakness_oracle`	**1.000**	(0.992, 1.000)	**1.000**
`weakness_data_inferred`	1.000	(0.992, 1.000)	1.000
`weakness_noisy_group`	1.000	(0.992, 1.000)	1.000
`random`	0.346	(0.306, 0.389)	0.380
`weakness_wrong_group`	0.002	(0.000, 0.011)	0.003
`train_loss`, `simplicity`, `compression`, `mdl_program`, `flatness_proxy`, `validation`	0.000	(0.000, 0.008)	0.000

`dihedral_reflection`

Selector	Invariant rate	Wilson 95% CI	Mean OOD
`weakness_oracle`	**1.000**	(0.992, 1.000)	**1.000**
`weakness_data_inferred`	1.000	(0.992, 1.000)	1.000
`weakness_noisy_group`	0.972	(0.954, 0.983)	0.976
`random`	0.340	(0.300, 0.383)	0.413
`weakness_wrong_group`	0.018	(0.009, 0.034)	0.119
classical baselines	0.000	(0.000, 0.008)	0.107

`color_permutation` (partial positive)

Selector	Invariant rate	Wilson 95% CI	Mean OOD
`weakness_noisy_group`	0.858	(0.825, 0.886)	0.890
`weakness_oracle`	0.824	(0.788, 0.855)	0.875
`random`	0.322	(0.283, 0.364)	0.455
`weakness_data_inferred`	0.138	(0.111, 0.171)	0.325
classical baselines, `weakness_wrong_group`	0.000	(0.000, 0.008)	0.210

`parity_coset` (negative case)

Selector	Invariant rate	Mean OOD
`random`	0.298	0.341
`weakness_wrong_group`	0.038	0.038
`weakness_noisy_group`	0.022	0.022
`weakness_oracle`, `weakness_data_inferred`, classical baselines	0.000	0.000

Cyclic and dihedral exhibit the intended separation: the implemented generic baselines choose the local shortcut, while weakness chooses the invariant. The color-permutation result is a partial positive: even the oracle S_n score is informative, but the data-inferred cyclic prior is not enough to recover the hidden permutation structure. Parity is a clean negative: |G| = 2 is too small to separate the truth from train-perfect local patches that are also parity-compatible.

4. Neural Weakness as a Predictor of OOD Accuracy

4.1 Setup

We train small MLPs on cyclic-prefix-shift tasks with modulus 11 and training window 3. Hyperparameters vary across:

- depth $\in$ 1,2,3, hidden width $\in$ 16,32,64,128, initialization scale $\in$ 0.3,0.7,1.0,1.5;
- optimizer $\in$ Adam, SGD+momentum, learning rate $\in$ 10^-3, 3·10^-3, 10^-2, 3·10^-2, weight decay $\in$ 0,10^-4,10^-2;
- augmentation $\in$ {none, partial cyclic, full cyclic, wrong reflection, wrong random}.

After 2000 steps, we extract the full argmax function table hat f and compute training loss, held-out validation accuracy on observed-support examples, parameter L_2, Hutchinson sharpness proxy, true-group weakness, partial-cyclic weakness, wrong-random-permutation weakness, and random-label weakness. The dependent variable is OOD accuracy on the held-out suffix.

4.2 Local 256-MLP Sweep

Across 256 trained MLPs, mean OOD accuracy is 0.3335 and 23.8% of models reach perfect OOD accuracy.

Predictor	Pearson r	Spearman rho
`weakness_oracle_norm`	**+0.817**	**+0.552**
`weakness_partial_cyclic_norm`	+0.804	+0.540
`weakness_oracle` (raw)	+0.763	+0.715
Hutchinson sharpness proxy	+0.129	+0.142
parameter L_2	+0.099	+0.308
held-out validation accuracy	+0.096	+0.058
training loss	-0.031	+0.136
`weakness_wrong_group_norm`	**-0.129**	-0.057
`weakness_random_label_norm`	**-0.116**	-0.051

Per-augmentation breakdown:

Augmentation	n	Mean OOD	Mean weakness
`full_cyclic`	54	0.939	0.951
`partial_cyclic`	48	0.618	0.320
`wrong_random`	50	0.084	0.140
`wrong_reflection`	50	0.017	0.117
`none`	54	0.000	0.141

4.3 Modal Replications

The 1024-model Modal run uses 8 shards by 128 models with the same hyperparameter schema as the local sweep. The 4096-model rescale uses 32 shards by 128 models with base seed 20260702. These are larger stochastic draws from the same sweep family rather than new task families; the 4096 run is the highest-powered estimate.

Predictor	256 Pearson	1024 Pearson	4096 Pearson	4096 Spearman
`weakness_oracle_norm`	**+0.817**	**+0.813**	**+0.8085**	**+0.5417**
`weakness_partial_cyclic_norm`	+0.804	+0.804	+0.7940	+0.5343
parameter L_2	+0.099	+0.273	+0.2533	+0.3313
held-out validation accuracy	+0.096	+0.089	+0.0924	+0.0427
Hutchinson sharpness proxy	+0.129	+0.134	+0.0848	+0.1617
training loss	-0.031	-0.048	-0.0249	+0.1200
`weakness_wrong_group_norm`	-0.129	-0.116	-0.1040	-0.0218

Fisher-z 95% intervals on the 4096-model run are:

Predictor	Pearson r	95% CI
`weakness_oracle_norm`	**+0.8085**	**(+0.7976, +0.8189)**
`weakness_partial_cyclic_norm`	+0.7940	(+0.7824, +0.8051)
parameter L_2	+0.2533	(+0.2244, +0.2818)
held-out validation accuracy	+0.0924	(+0.0620, +0.1227)
Hutchinson sharpness proxy	+0.0848	(+0.0543, +0.1151)
training loss	-0.0249	(-0.0555, +0.0057)
`weakness_wrong_group_norm`	-0.1040	(-0.1342, -0.0736)

4096-model per-augmentation breakdown:

Augmentation	n	Mean OOD	Mean weakness
`full_cyclic`	805	0.9434	0.9545
`partial_cyclic`	806	0.6438	0.3678
`wrong_random`	841	0.0877	0.1230
`wrong_reflection`	823	0.0135	0.1315
`none`	821	0.0000	0.1157

Because augmentation condition raises both true-group weakness and OOD accuracy, we also run an augmentation fixed-effect check on the 4096 raw records. Residualizing both OOD accuracy and `weakness_oracle_norm` by augmentation condition leaves Pearson r = +0.4883 (Fisher 95% CI +0.4647 to +0.5113). Equivalently, the fixed-effect regression

```
OOD_i = alpha_augmentation(i) + beta W_G(hat f_i) + epsilon_i
```

gives beta = +0.3652 (95% CI +0.3452 to +0.3853). Within-condition correlations are positive in the informative `partial_cyclic` and `full_cyclic` regimes (r = +0.6242 and +0.4716 respectively) and near zero in regimes where OOD accuracy is almost degenerate (`none`, `wrong_random`, `wrong_reflection`). The headline correlation is therefore partly augmentation-mediated, but a substantial learned-function signal remains after controlling for augmentation labels.

The wrong-group and random-label controls rule out the trivial explanation that any equivariance count predicts OOD. The observed signal is specific to the group aligned with the deployment shift.

5. Wrong-, Noisy-, and Data-Inferred-Group Ablations

The benchmark exposes a precise operating regime:

- **Wrong group**: random non-cyclic permutations select the invariant in only 0.2% of cyclic trials and 1.8% of dihedral trials.
- **Noisy group**: a transformation set containing roughly half the true group plus one wrong element still recovers the invariant in 100% of cyclic and 97.2% of dihedral trials.
- **Data-inferred group**: consistency filtering over the circular-domain prior recovers the invariant in 100% of cyclic and dihedral trials. This uses the domain's enumerable transformation prior and observed pair consistency, not the hidden offset, hidden reflection parameter, or ground-truth function.
- **Too small**: parity fails because the group is not rich enough to separate the truth from local patches.
- **Too large/unstructured**: S_n is informative for the oracle selector but difficult for the current data-inferred heuristic; wrong involutions can have comparable centralizer-orbit sizes.

These failures matter. Weakness is not a universal scalar that solves OOD generalization. It becomes load-bearing when the candidate transformation set is aligned with the deployment-generating structure and has enough resolution to distinguish train-perfect completions.

6. Scaling Beyond Cyclic Symbolic Tasks

6.1 Vision: Rotated-Stroke Partial-Orbit Supervision

We construct a synthetic stroke-classification task with eight classes on 16 by 16 grayscale images. The underlying transformation is Z_8 rotation. Training shows each class at only three of the eight rotation angles; the remaining angles are OOD. This mirrors Perin and Deny's partially observed cyclic-symmetry setup, with neural weakness added as a measurement.

Across 96 CNN/MLP models:

Predictor	Pearson r	Spearman rho
`weakness_rotation_norm`	**+0.672**	**+0.573**
`weakness_wrong_group_norm`	**-0.341**	**-0.452**

parameter L_2	+0.333	+0.269
train accuracy	+0.332	+0.265
final training loss	-0.319	-0.221
Hutchinson sharpness	+0.198	+0.368

Per-augmentation breakdown:

Augmentation	n	Mean OOD	Mean weakness
`partial_rotation`	25	0.734	0.776
`wrong_permute`	20	0.264	0.388
`none`	28	0.275	0.402
`full_rotation`	23	0.272	0.432

The wrong-group control is anti-correlated, consistent with a real tradeoff between rotation-compatible structure and pixel-permutation-compatible structure.

Appendix C reports an exploratory Pythia-70M latent-geometry check. It is not used as evidence for OOD behavioral generalization.

7. Related Work

- **Shortcut learning and underspecification.** Geirhos et al. frame shortcuts as rules that solve benchmarks but fail under changed test conditions. D'Amour et al. show that standard held-out validation can leave pipelines underspecified, returning predictors with similar validation performance but different deployment behavior. Our benchmark is a controlled finite-domain version of this ambiguity.
- **Domain generalization and invariant prediction.** Invariant Risk Minimization seeks representations whose optimal classifier is stable across training environments, and domain-generalization surveys organize a broad literature on OOD generalization without target-domain labels. Our setting is narrower: we do not learn invariances from multiple natural environments, but score train-consistent finite functions by compatibility with a candidate deployment-generating transformation family.
- **Geometric deep learning and equivariance.** Cohen and Welling, Kondor and Trivedi, and Bronstein et al. formalize how architectural equivariance builds transformation structure into the model class. Our contribution is a measurement/selection criterion for candidate functions and learned function tables, rather than a new equivariant architecture.
- **Symmetry learning from data.** Perin and Deny analyze partially observed cyclic orbits and show when conventional supervised networks fail to generalize symmetries not present in the architecture or sufficiently dense in the data. Van der Ouderaa, Immer, and van der Wilk learn layer-wise equivariances using gradients and model evidence. Our data-inferred selector is much simpler and enumerative, but it targets the same gap: useful symmetries are often not known in advance.
- **Simplicity, MDL, and function-space priors.** Solomonoff and Hutter provide the algorithmic-probability foundation; Rissanen and Grunwald formalize MDL as a model-selection principle; Valle-Perez, Camargo, and Louis and Mingard et al. argue that neural parameter-function maps are biased toward simple functions. Our result is not a refutation of simplicity bias. It shows that, in these

shortcut-compatible constructions, the tested generic simplicity proxies prefer the wrong train-perfect completion.

- **Flatness and sharpness.** Hochreiter and Schmidhuber and Keskar et al. motivate flat minima; Dinh et al. show that sharpness can be changed by reparameterization without changing the represented function. Bennett's recent flat-minima critique argues for weakness as a function-level alternative. Our Hutchinson proxy is only one flatness measure, so the empirical claim is deliberately narrow.
- **Grokking and emergence of structure.** Power et al. and Liu et al. document transitions from memorization to structured generalization. The neural sweeps here give one concrete variable, learned-function weakness, that tracks the structured side of that transition in cyclic tasks.

8. Limitations and Negative Results

1. The symbolic benchmark uses small finite domains. Exact weakness counting will not scale to large or continuous groups without generator-based approximations or sampling.
2. The data-inferred selector is not open-ended group discovery. It uses an enumerable transformation prior for the domain and filters it by training-pair consistency.
3. The MDL, compression, validation, norm, and flatness baselines are intentionally simple proxies. Stronger PAC-Bayes, compression, or validation schemes could be tested and should be.
4. The neural tasks are synthetic and small. The 4096-model rescale improves statistical stability but does not make the cyclic task naturalistic.
5. The vision task is still synthetic, although it is closer to standard partial-orbit image settings than the symbolic benchmark.
6. The language result is latent-geometric only. It should not be interpreted as behavioral paraphrase robustness in LLMs.
7. Weakness fails when the transformation set is too small, too large, or not aligned with the deployment shift. Those failures are part of the theory's boundary, not exceptions to hide.

9. Reproducibility

All core code is in `experiments/symbolic_weakness/`. The committed result reports are:

- `experiments/symbolic_weakness/results/multi_family_500_2026_06_09.md`
- `experiments/symbolic_weakness/results/neural_sweep_v3_2026_06_09.md`
- `experiments/symbolic_weakness/results/modal_neural_sweep_v1_2026_06_09.md`
- `experiments/symbolic_weakness/results/modal_neural_sweep_2026_07_02.md`
- `experiments/symbolic_weakness/results/neural_correlation_checks_2026_07_02.md`
- `experiments/rotation_weakness/results/sweep_v1_2026_06_09.md`
- `experiments/paraphrase_weakness/results/pythia_70m_2026_06_09.md`

Commands:

```
# Symbolic multi-family benchmark
python3 -m experiments.symbolic_weakness.benchmark \
    --trials-per-family 500 --seed 20260609 \
    --out artifacts/symbolic_weakness/multi_family_500.json

# Local neural sweep
python3 -m experiments.symbolic_weakness.neural \
    --n-models 256 --epochs 2000 --base-seed 20260609 \
    --out artifacts/symbolic_weakness/neural_sweep_v3.json

# 1024-model Modal neural sweep
doppler --scope /Users/jawaun/superoptimizers run -- \
```

```
    uvx --python 3.12 --from modal modal run \
    experiments/symbolic_weakness/modal_neural_sweep.py \
    --n-shards 8 --models-per-shard 128 --epochs 2000 \
    --base-seed 20260609 \
    --out artifacts/symbolic_weakness/modal_neural_sweep_v1.json

# 4096-model Modal rescale
doppler --scope /Users/jawaun/superoptimizers run -- \
    uvx --python 3.12 --from modal modal run \
    experiments/symbolic_weakness/modal_neural_sweep.py \
    --n-shards 32 --models-per-shard 128 --epochs 2000 \
    --base-seed 20260702 \
    --out artifacts/symbolic_weakness/modal_neural_sweep_2026_07_02.json
```

Quality gates:

```
python3 -m unittest discover -s tests -p "test_symbolic*"
python3 -m unittest discover -s tests -p "test_rotation*"
python3 scripts/regen.py symbolic_weakness
```

10. Discussion

The discriminating quantity in these experiments is not generic shortness alone. It is compatibility with the transformations that generate the missing cases. When a local shortcut and a global invariant are both train-perfect, loss and in-distribution validation are silent; the tested simplicity proxies prefer the shorter local patch; and parameter-space diagnostics are at best indirect. Weakness directly scores the structure that matters for the OOD split.

The data-inferred result is the practical version of the symbolic claim. The selector does not need the hidden offset or hidden reflection; it needs an enumerable transformation prior and enough observed pair structure to avoid contradiction. This is still far from open-ended symmetry discovery, but it is a meaningful step away from oracle-group scoring.

The negative cases should guide future work. Parity says that some transformation sets are too coarse to select the transportable rule. The S_n case says that a huge, weakly structured group can make many wrong completions look similarly compatible. The language result says that latent clustering is not behavior. These boundaries make the claim more useful: weakness is a model-selection variable for regimes where the candidate transformation set is aligned, granular, and behaviorally relevant.

The next steps are clear: replace enumeration with learned transformation generators, test stronger compression and PAC-Bayes baselines, scale from synthetic groups to natural partial-orbit datasets, and turn weakness from a post-hoc selector into a training-time regularizer.

Appendix A. Symbolic Methods

A.1 Formal Task Definitions

Each symbolic trial is a finite supervised problem with domain X = 0,...,n-1, train set D_train, OOD input set X_ood, candidate pool mathcalF, invariant candidate f^star, and candidate transformation family G. All selector comparisons are restricted to train-perfect candidates.

Family	Domain and group	Truth	Training split	OOD split	Candidate pool
`cyclic_prefix_shift`	X=Z_n, n ∈ 7,11,13, G=Z_n cyclic shifts	f_b(x)=(x+b) mod n, b ∈ 1,...,n-1	prefix x ∈ 0,...,w-1, w ∈ 3,4,5	suffix x ∈ w,...,n-1	local prefix patch, memorizer, true shift, all wrong global shifts
`dihedral_reflection`	X=Z_n, n ∈ 7,9,11,13, G=D_n rotations plus reflections	f_b(x)=(b-x) mod n	prefix x ∈ 0,...,w-1	suffix x ∈ w,...,n-1	local patch then identity, memorizer, true reflection, train-consistent rotation shifts
`parity_coset`	even n ∈ 6,8,10, G=Z_2 parity swap	f(x)=x⊕ 1	one sampled parity coset	opposite parity coset	seen-coset patch, memorizer, parity swap, identity
`color_permutation`	X=0,...,n-1, n ∈ 4,5,6, G=S_n	sampled non-identity permutation π	sparse sampled subset of size w	unobserved inputs	local train patch, memorizer, true permutation, sampled train-consistent wrong permutations

The local-patch candidates copy every observed training output and use an identity/default value elsewhere. This makes them short and train-perfect but usually non-transportable. The invariant candidate is the true globally defined rule. Wrong invariant candidates are included when they also fit the observed pairs, so the candidate pool contains real ambiguity rather than a single obvious global rule.

The invariant-recovery criterion is whether the selector chooses the candidate named by `trial.invariant_name`. OOD accuracy is exact agreement with the invariant candidate on `trial.ood_inputs`.

A.2 Selector Definitions

All selectors operate on train-perfect candidates unless otherwise noted. "Uses labels" means the selector inspects observed training outputs; "uses group" means it scores against a transformation set.

Selector	Score optimized	Tie-break	Uses labels?	Uses group?

`train_loss`	observed training accuracy	lower `form_length`	yes	no
`validation`	leave-one-observed-pair-out accuracy	lower `form_length`	yes, in-distribution only	no
`simplicity`	lower hand-coded `form_length`	random among exact ties	no	no
`compression`	lower `form_length + 20 * train_errors`	random among exact ties	yes	no
`mdl_program`	higher 2^-form_length	lower `form_length`	no	no
`flatness_proxy`	larger count of domain positions unconstrained by training	lower `form_length`	yes	no
`weakness_oracle`	higher W_G(f) under the task group	lower compression length	no test labels; true candidate group	yes, oracle group
`weakness_wrong_group`	higher weakness under random non-cyclic permutations	lower compression length	no	yes, wrong control group
`weakness_noisy_group`	higher weakness under identity plus a random half-subset of G plus one wrong element	lower compression length	no	yes, noisy true-group subset
`weakness_data_inferred`	higher weakness under Ĝ inferred from training-pair consistency	lower compression length	yes, train pairs only	yes, inferred group
`random`	none	uniform random train-perfect candidate	no	no

The "classical baselines" column in the main tables aggregates `train_loss`, `validation`, `simplicity`, `compression`, `mdl_program`, and `flatness_proxy`. It should be read only as this implemented set of proxies.

A.3 Data-Inferred Group Algorithm

The data-inferred selector is not open-ended symmetry discovery. It assumes a small enumerable domain-action prior and filters that prior by training-pair consistency.

```
Input: training pairs D = {(x_i, y_i)}, domain size n, family tag

Choose base transformation prior T:
  cyclic_prefix_shift: cyclic shifts of Z_n
  dihedral_reflection: rotations and reflections of D_n
  parity_coset: identity and parity swap
  color_permutation: cyclic subgroup heuristic

Initialize G_hat = empty set.
For each input-side transformation g in T:
  Keep g if there exists an output-side transformation h in T such that
  every observed pair is non-contradictory:
      for all (x, y) in D, if g(x) is also observed with label y',
      then h(y) == y'.
Ensure identity is included.
Return G_hat.
```

No leakage: the algorithm uses only the observed training inputs and outputs, the domain size, and the enumerable transformation prior for the family. It does not use test labels, OOD inputs, the hidden shift offset b, the hidden reflection parameter, the sampled permutation π, or the identity of the invariant candidate.

Appendix B. Neural and Vision Methods

B.1 MLP Training Distribution and OOD Split

The neural cyclic benchmark samples finite cyclic-prefix-shift tasks and trains small MLPs for 2000 full-batch steps. The local 256-model sweep uses `experiments.symbolic_weakness.neural`; the Modal sweeps use `experiments.symbolic_weakness.modal_neural_sweep`.

For the Modal runs, each model samples:

- modulus n ∈ 7,11,13;
- train window w ∈ 2,3,4, with OOD inputs w,...,n-1;
- augmentation in {`none`, `partial_cyclic`, `full_cyclic`, `wrong_reflection`, `wrong_random`};
- augmentation count in 2,4,6,8 for partial/wrong augmentations and 0 for none/full;
- hidden width in 16,32,64,128, depth in 1,2,3, initialization scale in 0.3,0.7,1.0,1.5;
- optimizer in {Adam, SGD with momentum}, learning rate in 10^-3,3·10^-3,10^-2,3·10^-2, and weight decay in 0,10^-4,10^-2.

After training, the model is converted to a full argmax function table over the finite domain. Weakness scores are computed on that learned function table, not on hidden activations.

B.2 Correlation Intervals and Augmentation Fixed Effects

Correlation intervals use the Fisher transform: `z = atanh(r)`, `SE(z) = 1 / sqrt(n - 3)`, and `CI_95 = tanh(z +/- 1.959963984540054 * SE(z))`.

Selected Pearson intervals:

Run	Predictor	r	Fisher 95% CI

256 local	`weakness_oracle_norm`	+0.8169	(+0.7716, +0.8540)
256 local	`weakness_partial_cyclic_norm`	+0.8035	(+0.7553, +0.8431)
256 local	held-out validation	+0.0960	(-0.0269, +0.2161)
256 local	parameter L_2	+0.0991	(-0.0238, +0.2190)
256 local	Hutchinson sharpness	+0.1294	(+0.0069, +0.2481)
1024 Modal	`weakness_oracle_norm`	+0.8132	(+0.7914, +0.8330)
1024 Modal	parameter L_2	+0.2731	(+0.2154, +0.3289)
4096 Modal	`weakness_oracle_norm`	+0.8085	(+0.7976, +0.8189)
4096 Modal	`weakness_partial_cyclic_norm`	+0.7940	(+0.7824, +0.8051)
4096 Modal	held-out validation	+0.0924	(+0.0620, +0.1227)
4096 Modal	parameter L_2	+0.2533	(+0.2244, +0.2818)
4096 Modal	Hutchinson sharpness	+0.0848	(+0.0543, +0.1151)
4096 Modal	training loss	-0.0249	(-0.0555, +0.0057)
4096 Modal	wrong-group weakness	-0.1040	(-0.1342, -0.0736)

For the augmentation fixed-effect check, we subtract each augmentation condition's mean from both OOD accuracy and `weakness_oracle_norm`, then correlate the residuals. This is equivalent to fitting the one-predictor fixed-effect regression shown in Section 4.3. On the 4096-model raw records, residual r=+0.4883 and beta=+0.3652 show that weakness still predicts OOD variation inside augmentation strata, although the headline effect is partly mediated by augmentation condition.

B.3 Vision Task Generation

The rotated-stroke benchmark uses eight hand-coded stroke classes rendered on 16 by 16 grayscale grids. The transformation group is Z_8 rotation in 45-degree increments. For each class, the split samples 3 of 8 rotations for training and holds out the remaining 5 rotations for OOD evaluation. Each class-rotation cell is materialized with repeated noisy samples (`noise_std=0.05` by default). The sweep trains 96 small CNN/MLP models for 250 epochs with architecture, width, depth, initialization scale, optimizer, learning rate, weight decay, and augmentation sampled as reported in `experiments/rotation_weakness/results/sweep_v1_2026_06_09.md`.

Vision weakness is the fraction of sample/rotation pairs for which the model's argmax prediction is invariant under rotating the input. The wrong-group control repeats the computation under random pixel permutations of equal cardinality.

Appendix C. Exploratory Pythia-70M Latent Geometry

We extract per-layer mean-pooled hidden states from Pythia-70M for 24 concepts with 3 paraphrase variants each, treating each concept's variants as an approximate paraphrase orbit. We compute same-orbit cosine, wrong-orbit cosine, and next-token argmax consistency. Raw cosine is dominated by embedding anisotropy, so we also apply per-layer mean-centering following All-but-the-Top.

After centering, same-orbit paraphrase similarity exceeds the wrong-orbit control by +0.44 to +0.79 across layers:

Layer	Same-orbit centered	Wrong-orbit centered	Gap
0	0.400	-0.038	+0.438
1	0.568	-0.020	+0.588
2	0.565	-0.030	+0.596
3	0.718	+0.029	+0.689
4	0.742	+0.008	+0.734
5	0.726	-0.067	**+0.793**
6	0.700	-0.014	+0.713

This is a real latent-geometry signal, but it does not predict per-concept next-token behavioral consistency at this scale: Pearson(centered weakness, behavior) lies in [-0.35, -0.13] across layers with only 24 concepts. We therefore treat it as exploratory. It supports "paraphrase orbits cluster in centered latent space," not "latent weakness predicts LLM behavior."

Acknowledgements and Code

The benchmark, neural sweep, Modal entrypoint, summarization tools, and tests are available at https://github.com/jawauntb/research-derived-experiments under `experiments/symbolic_weakness/`.

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